

Project Outline: IE500 Data Mining

Ethnic Bias and Generalization in Facial Attractiveness Prediction

Team Number: 3

Maxim Froehlich, David Siregar, Timon Kuhl,
Lars Bullmahn, Jagadeesh Gunti, Simon Heinz

April 12, 2026

Submitted to

Prof. Dr. Christian Bizer
University of Mannheim

1 Problem Description

Automated systems for rating or ranking human faces are increasingly used in applications ranging from online platforms to hiring tools. Unlike many computer vision tasks, facial attractiveness prediction is inherently subjective and culturally shaped, which makes it particularly prone to encoding demographic bias. Models trained on data from a narrow set of ethnic groups may learn cultural beauty standards rather than broadly applicable visual patterns, raising both methodological and ethical concerns. We study this problem as a supervised regression task, where the goal is to predict the mean attractiveness rating assigned by human annotators to a given face image. We combine three datasets: SCUT-FBP5500 [1] (Asian/Caucasian), MEBeauty [2] (six ethnic groups), and LiveBeauty [3] (large-scale Asian), to examine cross-dataset generalization and ethnic bias.

Our core research questions are: (1) How well do models generalize across datasets with different ethnic compositions, and does training on more ethnically diverse data reduce per-group performance disparities? (2) Which geometric facial measurements drive attractiveness predictions across ethnic groups, and how well do interpretable landmark-based features perform relative to end-to-end deep models?

2 Data Acquisition

We use three publicly available datasets. **SCUT-FBP5500** [1] contains 5,500 face images (Asian and Caucasian subjects) with attractiveness ratings from 60 annotators per image on a 1–5 scale, available on Kaggle. **MEBeauty** [2] contains 2,550 images across six ethnic groups (Asian, Black, Caucasian, Hispanic, Indian, Middle Eastern) with ratings on a 1–10 scale, available on GitHub (fbplab/MEBeauty-database). **LiveBeauty** [3] contains approximately 10,000 Asian face images with around 200,000 attractiveness labels, providing a large-scale single-ethnicity corpus that complements SCUT-FBP5500’s Asian subset and strengthens cross-dataset generalization analysis. We will verify completeness (expected missing rate <1%) and check for corrupt files.

For cross-dataset evaluation, we treat the three datasets as separate corpora: models trained on one are tested on the others to quantify generalization failure. To make scores comparable across datasets, we apply z-score normalization per dataset (zero mean, unit variance). Within MEBeauty, group-stratified splits enable per-ethnicity performance breakdowns.

3 Methodology

3.1 Preprocessing

- **Face Detection & Alignment:** Standard face detection crops and aligns all images to a consistent resolution across both datasets.
- **Feature Extraction:** Interpretable tabular features computed from MediaPipe facial landmarks serve as the primary input representation. The current feature set includes geometric measurements like interpupillary ratio, eye spacing ratio or mouth-to-chin ratio. Additional geometric features may be engineered based on initial model performance and feature importance analysis.
- **Normalization:** Rating scores are z-score normalized per dataset (zero mean, unit variance) so that all three datasets can be combined for joint training. Tabular geometric features are standardized using z-scores.
- **Outlier Handling:** Per-image annotator standard deviation is computed from individual rater scores. Images in the top 10% of disagreement within each dataset are flagged as high-variance; we report metrics with and without these outliers.

3.2 Proposed Algorithms

Baseline: Global mean predictor; Ridge Regression (α tuned via Optuna) on tabular geometric features.

Symbolic: Decision Tree Regressor (max depth $\in \{3, 5, 7, 10\}$, min samples leaf $\in \{5, 10, 20\}$), which enables inspection of which geometric features drive predictions and identification of interpretable decision rules.

Ensemble: Random Forest Regressor ($n_estimators \in \{100, 200, 500\}$, max features $\in \{\text{sqrt}, \log_2\}$) with feature importance analysis to identify which facial measurements are most predictive of attractiveness across different ethnic groups.

Boosting: XGBoost, LightGBM, and CatBoost regressors with tuned tree depth, learning rate, and number of estimators to capture nonlinear interactions between geometric features while often outperforming a single tree or bagged ensemble on tabular data.

Stacking Ensemble: StackingRegressor combining XGBoost, Random Forest, Gradient Boosting, and Ridge as base learners with a Ridge meta-learner, providing a committee-based prediction that mitigates individual model biases.

Neural Baseline: A three-layer MLP (128-64-32 units, ReLU activations) trained on tabular features with Adam ($lr = 0.001$), batch size 64, up to 500 iterations with early stopping (patience 30) on validation loss.

Data is split into 80% training / 20% held-out test, stratified by ethnicity. Within the training data, hyperparameters are tuned via Optuna (TPE sampler, 200 trials) Bayesian optimization with stratified 5-fold cross-validation. Final evaluation is reported on the held-out test set. For cross-dataset evaluation, models are trained on one dataset and tested directly on the others. Feature importance analysis will guide potential expansion of the geometric feature set.

4 Evaluation and Expected Results

4.1 Measuring Success

Regression metrics: MSE/RMSE (primary), Pearson Correlation Coefficient (PCC), and MAE.

Generalization: Train on SCUT, test on MEBeauty (and vice versa). The performance drop quantifies cross-ethnic generalization failure.

Fairness / Bias: Fairness is operationalized as consistency of predictive performance across demographic groups. We compute per-ethnicity RMSE within MEBeauty for each model and summarize disparity using: (1) the absolute gap between worst-group and best-group RMSE, (2) the ratio of worst-to-best RMSE (values closer to 1 indicate fairer performance), and (3) the standard deviation of per-group RMSEs. These metrics quantify whether the model systematically underperforms for specific ethnic groups, which would indicate learned bias toward certain demographic features or beauty standards.

State-of-the-art comparison: Results benchmarked against published baselines for SCUT-FBP5500 (prior CNN approaches achieve $PCC \approx 0.90$) and related literature using end-to-end CNN or transformer architectures. These serve as external comparison points rather than models we necessarily reimplement.

4.2 Expected Outcomes

We expect tabular geometric features to provide interpretable and computationally efficient predictions, while showing notable cross-dataset performance drops when training on SCUT and testing on MEBeauty. We anticipate higher errors for underrepresented groups (Black, Hispanic, Middle Eastern) in SCUT-trained models and partial mitigation when training on MEBeauty. Feature importance analysis from tree-based models will reveal which geometric measurements (e.g., eye ratios, symmetry, proportions) are most predictive and whether these patterns differ across ethnic groups, exposing potential sources of cultural bias. Comparison with published CNN/transformer results will position the performance-interpretability tradeoff of our tabular approach.

Bibliography

- [1] L. Liang, L. Lin, L. Jin, D. Xie, and M. Li, “SCUT-FBP5500: A diverse benchmark dataset for multi-paradigm facial beauty prediction,” in *Proceedings of the 24th International Conference on Pattern Recognition (ICPR)*. IEEE, 2018, pp. 1598–1603.
- [2] I. Lebedeva, Y. Guo, and F. Ying, “MEBeauty: a multi-ethnic facial beauty dataset in-the-wild,” *Neural Computing and Applications*, pp. 1–15, 2021.
- [3] H. Li, X. Ren, H. Yu, Y. Chen, K. Li, L. Wang, X. Min, H. Duan, G. Zhai, and X. Liu, “FPEM: Face prior enhanced facial attractiveness prediction for live videos with face retouching,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2025, pp. 11 458–11 468.